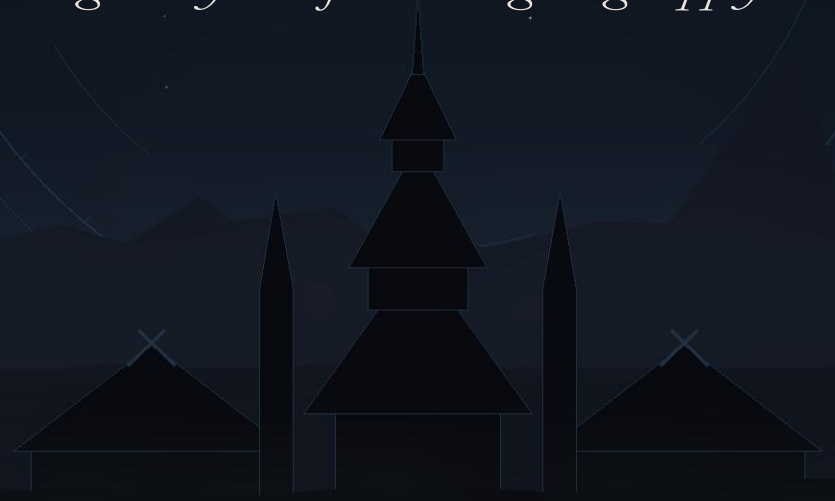
RAGNAROK

TRIAL OF THE NINE REALMS

An agentic intelligence system for investigating supply-chain disruption



AGENTIC AI HACKATHON · RAGNAROK

TEAM VIKSIT CODERS

A disrupted corridor can become a national risk before the picture is clear

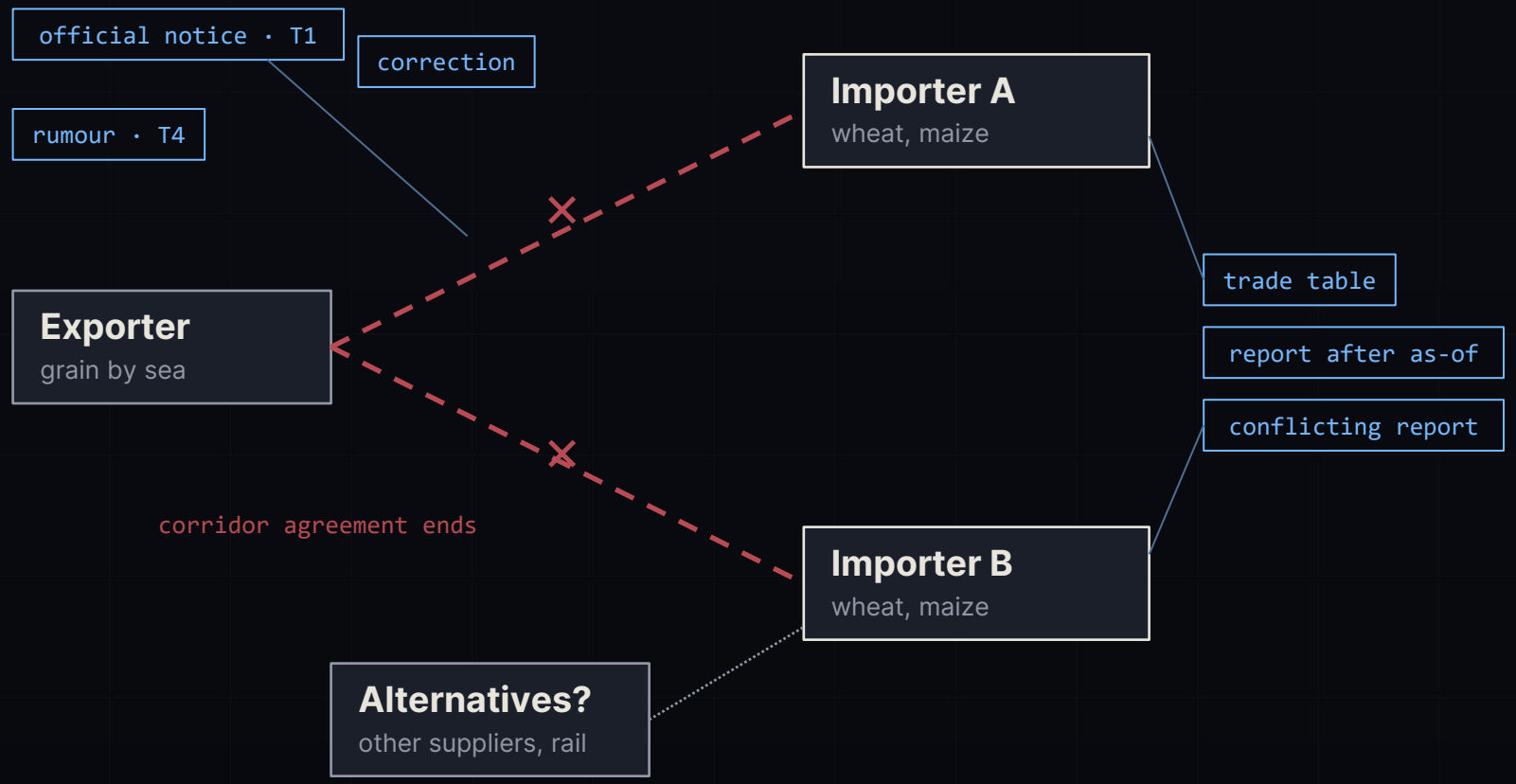


Illustration. Roles taken from the Runemark dev pack (fictional).

QUESTIONS THAT ARRIVE AT ONCE

- Which countries and sectors are exposed?
- How much trade is actually at risk?
- Can other routes, suppliers or stocks cover it?
- Which claims rest on reliable evidence?
- What is still missing?

The hard part is not finding information. It is reconciling it into a defensible assessment.

THE INSIGHT

One answer can hide an entire side of the story

ONE ANSWER



- Incomplete evidence
- One-sided interpretation
- Numbers no one can check

RESEARCH — Heimdall builds the case file and names what is missing

CHALLENGE — Thor and Loki argue opposite sides and attack each other's claims

VERIFY — Code checks citations and does every calculation

JUDGE — Odin rules on each claim and decides if more research is needed

A single answer can be right. Separating the stages makes disagreement and evidence gaps visible instead of hidden.

Meet the four agents behind the investigation



HEIMDALL

THE INVESTIGATOR

- Searches documents and trade tables
- Verifies the event, builds the case file
- Flags corrections, rumours and gaps
- Runs targeted research on a send-back

THOR

HIGH EXPOSURE

- Argues exposure is high
- Cited claims, high estimate
- Makes the case for severity

LOKI

LIMITED EXPOSURE

- Argues independently for limited exposure
- Substitutes, other routes, stocks
- Challenges overstated claims

ODIN

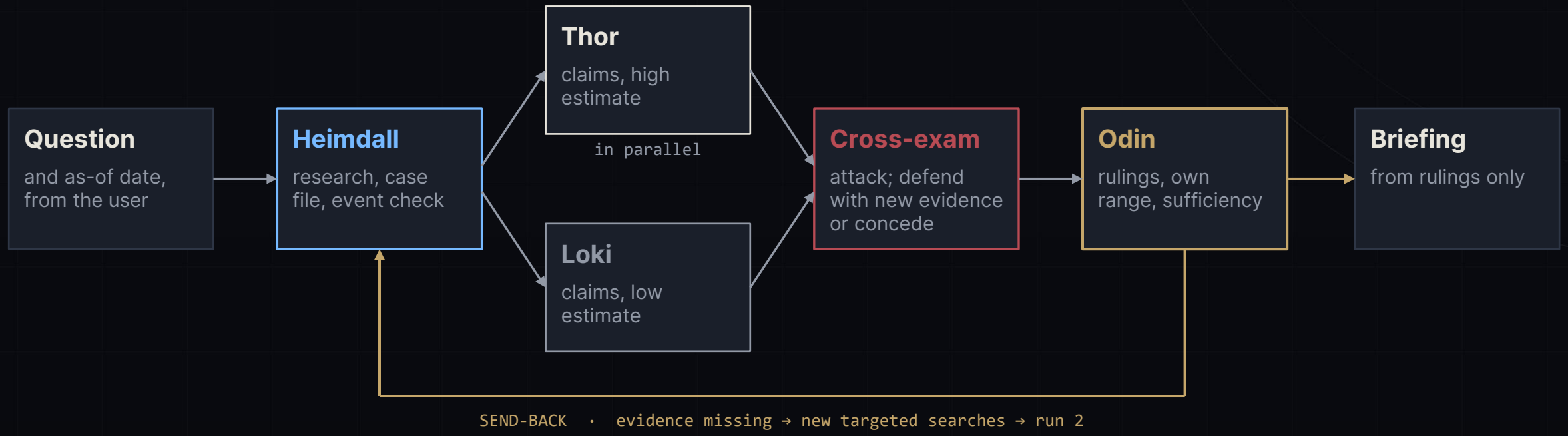
THE JUDGE

- Rules on every claim
- Builds his own range from evidence
- Decides if more research is needed
- Writes the briefing from his rulings

Heimdall opens every case. Odin has the last word. Thor and Loki see each other's claims only at cross-examination.

THE WORKFLOW

Not four chatbots. One orchestrated investigation.



FLOW IN CODE

An explicit state machine checks every transition. Agent output never picks the next step.

SHARED CASE FILE

Every agent works from Heimdall's case file and cites the same stable paragraph and row refs.

RECORDED DEBATE

Claims, attacks, defenses, concessions and rulings are append-only audit events.

BOUNDED LOOP

Repeated queries are refused; only new evidence counts. Runs are capped at 2 by default.

Every claim must survive scrutiny

CLAIM

Loki · L1-2 · inference

Midgard's maize imports from Orvan stayed more resilient after the agreement ended, thanks to delayed delivery terms.

ORV-0030 P14 | ORV-0030 R4

→

ATTACK

Thor · A1-T3

"Resilient" describes a delay. November value fell from USD 84.7M (2025) to 31.9M (2026).

ORV-0030 R11 | ORV-0030 R5

→

RESPONSE

Loki · concedes in part

"Resilient" inaccurately describes the maize trade. The claim is revised to a delayed but significant decline.

→

RULING

Odin · R-L1-2

Conceded. The proponent acknowledged the objection; the trade suffered heavy year-on-year losses.

FOUR VERDICTS, USED EVERYWHERE

- ● **established** — survived scrutiny on valid evidence
- ◐ **contested** — support, but credible counter-evidence
- ○ **conceded** — the claim's author withdrew it
- ■ **rejected** — Odin judged it unsupported

Conceded comes from the author; rejected from Odin. Every claim is tagged fact or inference. Real exchange from the saved case, condensed.

NUMBERS YOU CAN TRACE

The model reasons. Code does the arithmetic.

SOURCE DOCUMENT

ORV-0009
Trade note: Orvan to Midgard, wheat
tier 1 · 2027-01-19

→

CITED TRADE ROWS

Sep–Dec 2026
R3 R4 R5 R6

Sep–Dec 2025
R9 R10 R11 R12

→

DETERMINISTIC CALCULATION

CALC-2471c1 · yoy_change
sum(current) − sum(prior)
142.1 − 358.3 = −216.2 USD M

CALC-cb98e8 · Oct–Dec only
103.9 − 260.4 = −156.5 USD M

inputs kept · excluded rows: none

→

EXPOSURE RANGE

156.5 – 216.2
USD M · Midgard wheat

Odin's range: the year-on-year fall in import value

→

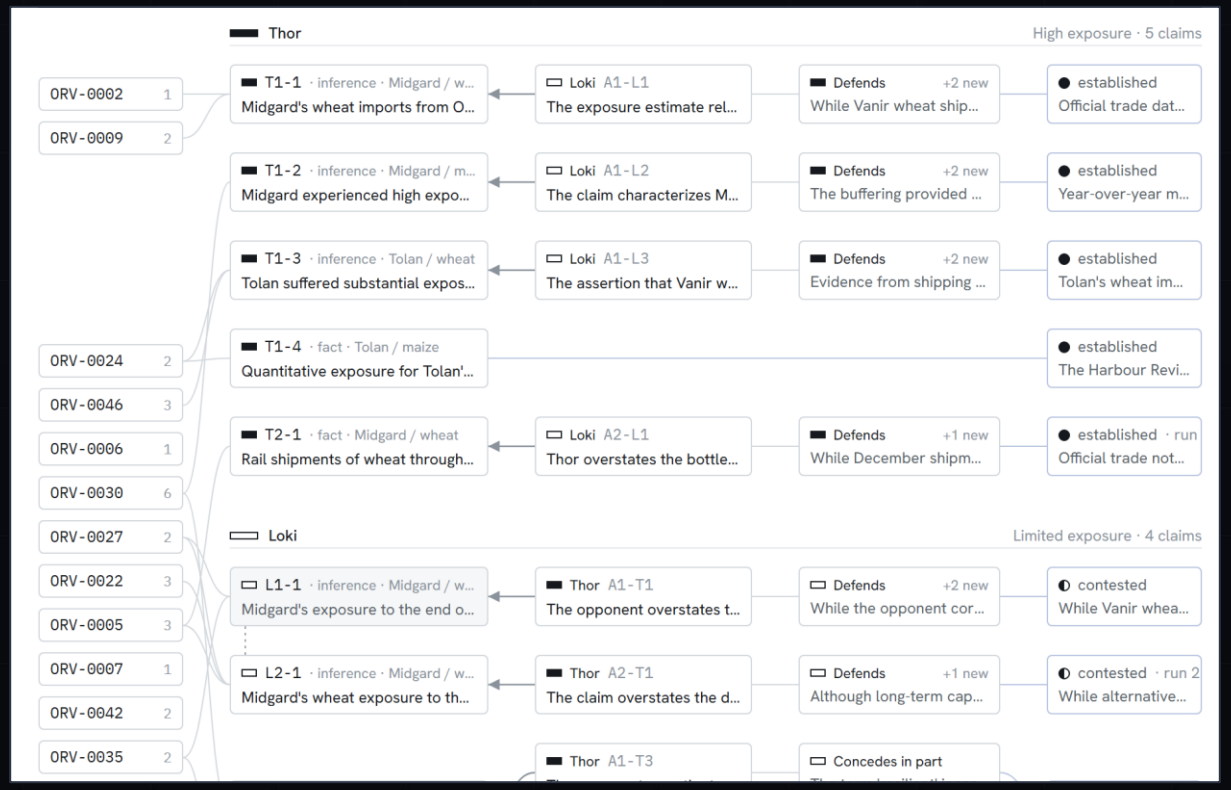
BRIEFING

Inline citation
[ORV-0009 R3, 2027-01-19]

- The model picks rows and an operation. Python computes the result and keeps the inputs.
- Each bound must equal a cited calculation result, and both bounds must be the same kind of calculation.
- Trade value at risk is not economic loss. A missing value is never treated as zero.

Every estimate states metric, unit, period, denominator, assumptions, missing data and confidence. Figures from case-20261026-005913-e4e654.

From investigation to decision-ready intelligence



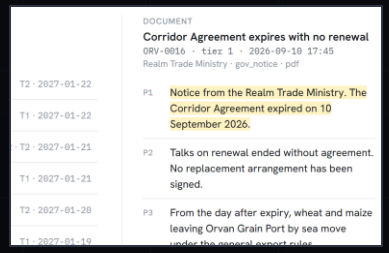
ARGUMENT MAP

Cited documents → claims → attacks → responses → Odin's rulings, one lane per claim



EXPOSURE

Thor, Loki and Odin on one axis; missing is never drawn as zero



CLICK-THROUGH

Any citation opens the source with the cited paragraph highlighted.

Real captures, saved two-run case.

The system must know what it cannot safely use

DOCUMENTS

PDF · TXT · DOCX, packs and uploads

VALIDATED INGEST

Type and size checked, safe file names. The original is kept with its content hash. Damaged files and rows are flagged, never used as valid numbers.

INJECTION SCAN

Role labels and override phrases are flagged. The span stays on record but is withheld from agent prompts.

TRAP_INJECTION

AS-OF GATE

Enforced in code on search, reads, trade rows, calculations and citations. A later document is refused (HTTP 403).

CLOCK_VIOLATION_BLOCKED

CITATION CHECK

At briefing build: the paragraph or row exists, the quote matches, the date is in the window. Failures are dropped and listed.

AGENTS

see only wrapped evidence

TRANSPARENT UNCERTAINTY

No estimate is shown as missing, never as zero. A briefing is labelled complete, partial or incomplete; it never claims completeness falsely.

EVERYTHING ON THE RECORD

Append-only audit log in SQLite with JSONL export. Corrections, retractions and duplicates stay linked to the reports they affect.

TECHNICAL ARCHITECTURE

A transparent application around a specialised agent workflow

FRONTEND	**Dashboard** React · Vite · TypeScript · Tailwind	**Six views** Trial · Argument map · Exposure · Evidence · Briefing · Run trace	**Case state** typed API client, live event stream
API	**FastAPI on Uvicorn** POST /cases · GET /cases/{id} · /resume	**Live updates** /events Server-Sent Events, polling fallback	**Evidence access** /docs · /trade · /feed · POST /ingest
AGENT LAYER	**Trial state machine** explicit stages, checked transitions, run limit	**Four agents** own prompt and tool allow-list; Pydantic-validated output, one repair	**LLM client** Gemini by default, Anthropic optional; model chain, budgets, reply cache
EVIDENCE AND AUDIT	**Ingest** PDF · TXT · DOCX, trade tables, damage and injection flags	**Hybrid retrieval** BM25 + bge-small vectors, RRF, metadata filters, as-of gate	**Calculations + store** deterministic records; SQLite audit log, memories and caches

Logical view. Runs locally as two processes (API on :8000, dashboard on :5173). 382 automated tests use a scripted model.

DEMONSTRATION

Follow one case from question to judgment



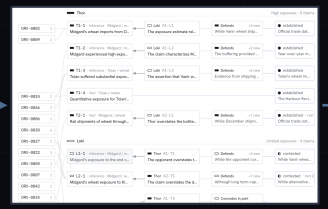
1 **Ask**

question and as-of date



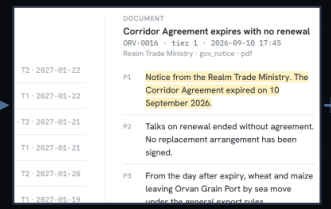
2 **Investigate**

Heimdall's case file



3 **Argue**

claims, attacks, rulings



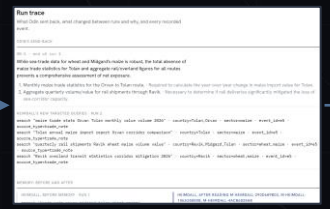
4 **Verify**

open a cited source



5 **Measure**

ranges and calculations



6 **Trace**

send-back, run 2, memory



7 **Brief**

report and exports

THE SAVED TWO-RUN CASE, A REAL GEMINI RUN ON THE DEV PACK

24	9	8	13	6
model requests	claims ruled	attacks, 4 each way	new references in run 2	logged memory changes

LIMITATIONS

- Conclusions are only as good as the evidence available.
- Cases can stay incomplete. This one did: no Tolan maize data is in the pack.
- Argument quality depends on the model; claims still need scrutiny.
- Daily model quotas can pause a run until the reset.
- A second run does not guarantee new evidence.


When the evidence is contested,
the decision must be earned.
Research the event. Challenge the claims. Trace the numbers. Judge the evidence.

Questions?

Endpoints, live updates and ingestion

ENDPOINT	PURPOSE
POST /cases	start a case (one at a time)
GET /cases/{id}	status and results
GET /cases/{id}/events	Server-Sent Events of the audit log
POST /cases/{id}/resume	resume a failed, partial or paused case
GET /docs/{id}[/{para}]	documents, as-of gated
GET /trade/{table}[/{row}]	trade rows, as-of gated
GET /feed?as_of=	documents with tier, correction, damage badges
POST /ingest	pack .zip or loose PDF, TXT, DOCX

LIVE UPDATES

The dashboard follows /events over Server-Sent Events and falls back to polling. Every visible action is a recorded event.

INGESTION FLOW

Manifest and header parse → text extraction with segment-wise encoding detection → [Pn] paragraphs and trade rows → damage, injection, duplicate and correction flags → hybrid index. New packs need no code changes.

OUTPUTS PER CASE

state, case file, claims, cross-examination, rulings, per-run folders, briefing (.md, .json), memory, model-request ledger, events.jsonl.

Why the investigation is split into stages

Separate research from argument

Heimdall establishes the facts and the gaps before anyone takes a side, so advocates start from the same case file.

Two opposing advocates

Thor and Loki search independently; the argument for limited exposure is never an afterthought.

Cross-examination with rules

Attacks need a target and counter-evidence or a named gap. Defenses need new evidence, or they concede.

A judge who cannot search

Odin rules from the record. He names missing evidence and can send the case back, but never assigns work.

Arithmetic in code

Replayable calculation records make every number in the briefing checkable.

Private memory per agent

General lessons carry across cases. The log shows the plan without memory and with it, so influence is visible.

NOT IMPLEMENTED · POSSIBLE NEXT STEPS

OCR for scanned PDFs (today they are ingested as empty, damaged documents) · non-English documents · Odin building ranges that differ from both advocates more often · more than one case at a time per API process